

SHUO LI LIU

Princeton Economics PhD Student | Decision Theory, Agentic Risk, Reference Alignment

Research focus: decision theory, agentic risk, reference alignment

Key interests: AI evaluation, governance, and the economics of knowledge work

Citizenship: Canadian

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Languages: English; Mandarin (fluent)

Coding: Python, MATLAB, Lean; LangGraph, PyTorch (LoRA), TensorFlow, pytest, Git

RESEARCH STATEMENT

I am a decision theorist and mechanism designer working on AI risk and governance. I study how advanced AI changes knowledge production, labor, and institutional capacity, combining economic theory, large-scale data analysis, and technical evaluation.

My research asks how institutions can evaluate AI systems when developers and agents have incentives to game their measures. I develop tools for assessing safety claims, tracking the coverage lost during verifier-supervised training, and aggregating judgments from panels of models. Four papers were accepted at ICML 2026 workshops: behavior-only deployment certification, model collapse despite verification, coherent preferences without expected utility, and incentive-compatible model-judge aggregation.

My empirical and policy research includes a study of 2.3 million Stack Overflow and Stack Exchange questions and work on Chinese innovation-finance policy with Richard B. Freeman. Across these projects, I connect formal assumptions and quantitative evidence to decisions about evaluation, oversight, and the economic effects of AI. I develop projects from theory through implementation and reproducible tests.

SELECTED PUBLICATIONS

- Grant, Liu & Yang (2024). “A Comment on: ‘Expected Uncertain Utility’.” *Econometrica*.
- Liu & Gasteratos (2019). “Assessing cell-based animal proteins.” *Science*.

EDUCATION

Princeton University — PhD student, Department of Economics (in progress). University Fellowship. Advisor: Faruk Gul. Fields: Economic Theory; Behavioral and Experimental Economics.

Harvard University — Master’s Degree in Finance, extension studies, 2017–2019. John Harvard Scholarship. Graduate Special Student (2018); Graduate School Associate Status (GSAS, 2019). Co-authored innovation-finance policy research with Richard B. Freeman ([Academy of Management Proceedings](#)).

McGill University — Bachelor’s Degree in Biochemistry and Mathematics, 2012–2016. James McGill Scholarship.

AI RISK RESEARCH — ACCEPTED

“Behavior-Only Deployment Certification Requires Bridge Assumptions.” *FAGEN @ ICML 2026, refereed poster*.

A safety certificate built only from what a model does cannot see past the cases you tested. Two models can agree on every test and still behave completely differently outside it. So certifying a model for deployment needs an extra, explicit assumption linking behavior to the mechanism underneath. I show this with a LoRA example. The result is about the limits of evaluation — not a claim that any live system is unsafe.

“Model Collapse Despite Ground-Truth Verification.” *ICML 2026 Workshop on Testing, refereed poster*.

If you keep training a model on its own outputs that a checker accepts — Lean-verified proofs, unit-test-passing code — the process behaves like genetic drift. The checked loss can sit at exactly zero while the model quietly loses coverage of what it was meant to learn. A warning for self-improvement loops: a checker confirms each step is locally valid, not that the target survives.

“Savage Without Monotonicity.” (with Jingni Yang) *PhilML @ ICML 2026, refereed poster*.

Drop one standard assumption — monotonicity — and coherent preferences no longer force expected-utility maximization. A foundations result about rationality, and about what we can and cannot assume when we call an advanced agent’s objective “coherent.”

“Bayesian Truth Serum for LLMs?” *AgenticUQ @ ICML 2026, refereed poster*.

I adapt the Bayesian truth serum — a mechanism for honest reporting — to pool judgments from panels of language-model judges. It rewards answers that are surprisingly common, not answers the crowd simply agrees on. Across a 406-panel benchmark, this cut calibration error from 0.0225 to 0.0102, and to 0.0077 in an information-only variant.

AI RISK RESEARCH — PORTFOLIO AND WORKING PAPERS

AI Labor Rubric — scoring multi-agent research systems by the value they actually add. A writer earns credit only when a judge confirms it fixed a weakness the red team found, so activity for its own sake counts for nothing. Given the same budget, the scored system reached 8/12, against 5/12 and 4/12 for baselines. *Under review: NeurIPS 2026 (Evaluations & Datasets, OpenReview [vTuyNeWXk](#)); ICML 2026 AI4Research workshop*.

Coverage Profiling Against Support Collapse in Agentic Drug Discovery. Agentic discovery pipelines can quietly lose coverage of biologically relevant targets while their headline metrics still look fine. This work measures that gap. It is an evaluation contribution, not a therapeutic one. *Under review: ICML 2026 GenBio workshop*.

PRF-Bench — RLHF Preference-Aggregation Audit. When you pool everyone’s preferences into one reward model, whom does it actually represent? On MultiPref, a pooled model hits the accuracy ceiling (69.6%) yet disagrees with the top choice of 47.7% of ordinary annotators and 55.4% of experts. The point is not that RLHF is broken — it is that aggregation is a question of whom the model speaks for. Public audits across SHP, OASST1, HH-RLHF, and MultiPref.

Constitutional AI as Judgment Aggregation. I model an AI constitution as a problem of combining many judgments — critic panels, interpretation, incentive-aware aggregation — using social choice and mechanism design. The main result is a trilemma: consistency when rules are combined, resistance to gaming, and honest feedback cannot all hold at once. Repository: [constitutional-trilemma](#).

Public Knowledge Labor after Generative AI (sole-authored; manuscript in preparation). What happens to human knowledge-sharing once a machine can answer the question? A study of 2.3 million Stack Overflow and Stack Exchange questions across 16 technical fields, 2020–2025. Using event studies and difference-in-differences around the ChatGPT release and Stack Overflow’s December 2022 AI-content policy, I track answer speed, resolution, accepted answers, who keeps contributing, and skill formation. It doubles as a public benchmark for whether AI grounds its claims in community-vetted evidence. Repositories: [who-still-answers](#), [stackoverflow-chatgpt-governance](#).

Agent Memory and Feedback Architecture Benchmark. A falsification benchmark for agent scaffolds that claim to have memory. Across 1,799 trials, calling the model over and over does not by itself create durable memory (about 1.3% semantic success).

DECISION THEORY PUBLICATIONS

Grant, S., Liu, S. L., and Yang, J. (2024). “A Comment on: ‘Expected Uncertain Utility’.” *Econometrica*, 92(1):247–256. [DOI](#).

Dong, X. and Liu, S. L. (2025). “Convexity under Ambiguity.” *Economic Theory Bulletin*, 13:353–373. [DOI](#).

TECHNOLOGY & INNOVATION POLICY

Innovation-finance policy (with Richard B. Freeman). Liu, M., Liu, S. L., Wang, Y., and Freeman, R. B. (2020). “Patent Collateralization: The Effect of Policy and Impact on Patenting.” *Academy of Management Proceedings* (peer-reviewed), 2020(1):20539. [DOI](#). A quantitative evaluation of Chinese innovation-finance policy, 2002–2015, using policy-timing variation across cities, with Richard B. Freeman (Harvard, NBER).

Technology & agricultural-development policy (UNDP). Lavoie, F., et al. (2018). *Agricultural Value Chains Analysis in Oe-Cusse, Timor-Leste*. UNDP Timor-Leste, 81 pp. (Editor: Shuo Li Liu). [PDF](#). A technology-capacity and value-chain assessment feeding development-policy recommendations.

Emerging-technology capacity assessment. Liu, S. L. and Gasteratos, K. (2019). “Assessing cell-based animal proteins.” *Science*, 363(6429):826. DOI. A quantitative technology-capacity assessment of cellular agriculture for global protein supply.

DECISION THEORY & FOUNDATIONS — WORKING PAPERS

Chang, Y. and Liu, S. L. (2024). *Counterexamples to Transitive Regret*. [arXiv:2407.00055](https://arxiv.org/abs/2407.00055). | Diecidue, He, Liu, and Somasundaram (2025). *Clarifications and Extensions to Regret Theory: A New Foundation*. SSRN 5356990. | Regret-theory series, SSRN [4148486](https://ssrn.com/abstract=4148486) / [4437540](https://ssrn.com/abstract=4437540) / [4437546](https://ssrn.com/abstract=4437546) / [4536728](https://ssrn.com/abstract=4536728).

RESEARCH INFRASTRUCTURE & SOFTWARE (SELF-BUILT, 2026)

Distributed AI Research Evaluation Pipeline. A 13-machine system running Claude Code and Codex. Per project it runs 17 writers, 17 red-team agents, and 5 judges under one rule: a writer is credited only when a judge confirms it fixed something the red team found — mechanism design applied to the pipeline itself. An LLM-rubric predictor acts as a quality gate before submission.

Multi-Agent Research Organization Simulator. A 137-agent, 12-department LangGraph pipeline, run through 19 full manuscript-revision rounds and used as workflow microdata.

Bayesian Truth-Serum Debate Aggregator. A formal incentive-compatibility proof backed by an 81-test suite. Repository: [truth-serum-debate](https://github.com/Truth-Serum-Debate). I have used Claude for Coding since 2024.

METHODS

Economic theory: axiomatic decision theory, mechanism design, information economics, social choice, behavioral and experimental choice, incentive-compatible elicitation. **AI evaluation:** adversarial benchmark design, support/coverage diagnostics, verifier and anti-hardcoding audits, panel aggregation, calibration/ECE, abstention accounting, failure taxonomies, JSONL transcript schemas. **Causal & empirical:** difference-in-differences, event studies, policy-timing designs, fixed-effects panels, pre-trend/placebo tests, wild-cluster bootstrap inference.

TALKS & SERVICE

Selected presentations: Foundations of Utility and Risk; Bay Area Experimental Economics; INFORMS Decision Analysis / Annual Meeting; HKS Seminar on International Economics. Memberships: Econometric Society; INFORMS. Harvard Math Question Center Teaching Assistant, 2019.

REFERENCES

Faruk Gul — Professor of Economics, Princeton University (PhD advisor).

Wolfgang Pesendorfer — Theodore A. Wells '29 Professor of Economics & Department Chair, Princeton University | pesendor@princeton.edu

Ryota Iijima — Professor of Economics, Princeton University | [rijima@princeton.edu](mailto:rrijima@princeton.edu)

Jingni Yang — Senior Lecturer, School of Economics, University of Sydney | jingni.yang@sydney.edu.au — co-author, *Econometrica* 2024 and *Savage Without Monotonicity* (PhilML @ ICML 2026).

Full contact details available on request.

SELECTED AI RESEARCH PROGRAMS & ASSESSMENTS

Incoming participant, September 2026 Iliad Intensive (mathematical AI alignment; LISA, London).

Advanced to Stage 4 (mentor-fellow matching), 2026-27 Winter PIBBSS Fellowship (August 2026).

Completed the Economics & Policy take-home assessment, Anthropic Fellows Program (June 2026).